

CR: HP047
LOAD\$ HP : BIOMETRIC THUMBPRINT
AUTHENTICATION REQUEST

Functional Requirement Specification (Addendum)

Version: 1.7

TABLE OF CONTENTS

1. INTRODUCTION	2
1.1. CR FORM	Error! Bookmark not defined.
1.2. SIGNOFF SOW	Error! Bookmark not defined.
2. ASSUMPTION	Error! Bookmark not defined.
3. OVERVIEW	3
4. NEW SCREEN	4
4.1. MENU FOR BIOMETRIC SCANNING	4
4.2. BIOMETRIC SEARCH SCREEN	5
4.2.1 <i>Field Definition for Biometric Search Screen</i>	5
4.2.2 <i>Search Application or Applicant/Seller for Biometric Scanning</i>	5
4.3. BIOMETRIC SCANNING DETAILS SCREEN	6
4.3.1 <i>Field Definition for Biometric Scanning Details Screen</i>	6
4.4. BIOMETRIC TAB (WITHIN APPLICATION)	7
4.4.1 <i>Field Definition for Biometric Tab</i>	7
4.5. BIOMETRIC VERIFICATION SCREEN	8
5. NEW MATRIX TABLES	9
5.1. MATRIX TABLE FOR BIOMETRIC SCANNING CHECKING CRITERIA	9
5.2. MATRIX TABLE FOR MANDATORY CHECK	12
6. NEW WORKFLOW VALIDATOR & VALIDATION LOGIC	13
6.1. VALIDATOR FOR MANDATORY CHECK	13
6.2. VALIDATOR FOR UNSUCCESSFULL BIOMETRIC	14
6.3. VALIDATION FOR BIOMETRIC MAXIMUM RETRY	14
6.4. VALIDATION FOR SUCCESSFULL BIOMETRIC	14
7. GENERAL CONSTANT	15
8. REPORT	16

1. INTRODUCTION

This document describes the detailed functional requirement specifications for Hire Purchase based on signed off Change Request SOW reference HLBB/LOAD\$/SOW/2015/072.

The purpose of this CR is to enhance the current Loan Origination System (LOAD\$) for HP. The document will cover the changes required to cater for Biometric Thumbprint Authentication Request.

2. OVERVIEW

Currently in LOAD\$, there is no functionality for biometric scanning. With this enhancement, users will be able to perform the biometric thumbprint checking for the applicant via LOAD\$ HP.

Changes required in LOAD\$ HP to accommodate this CR as follows:

1. System Integration with the Biometric Reader

LOAD\$ HP must be able to integrate with the following devices' API (application programming interface) to perform the biometric thumbprint verification:

(a) Sagem MorphoSmart 1350

(b) Dermalog ZF1+

Note: Confirmed by PMO on 11 Nov 2015 that this device integration is dropped from the delivery scope

(c) SmartTag (FID688, CSmarT680)

2. New Screens

LOAD\$ HP will introduce a new screen to perform Biometric Scanning and a new application tab for inquiry.

3. New Validator and Matrix Table

This validator is required to stop the application from proceeding if the biometric scanning has not yet completed. The validation will be based on the matrix table where the checking criteria will be defined by users.

4. New Constant

A new general constant is required to set the following:

(a) Attempt Counter (Number of tries the user can trigger the Biometric includes all status(include error))

(b) Mandatory Check (On/Off Control for the Validator)

(c) Successful Counter (Number of tries the user can trigger the Biometric does not include error. Only includes Matches/No Match)

5. Template Section

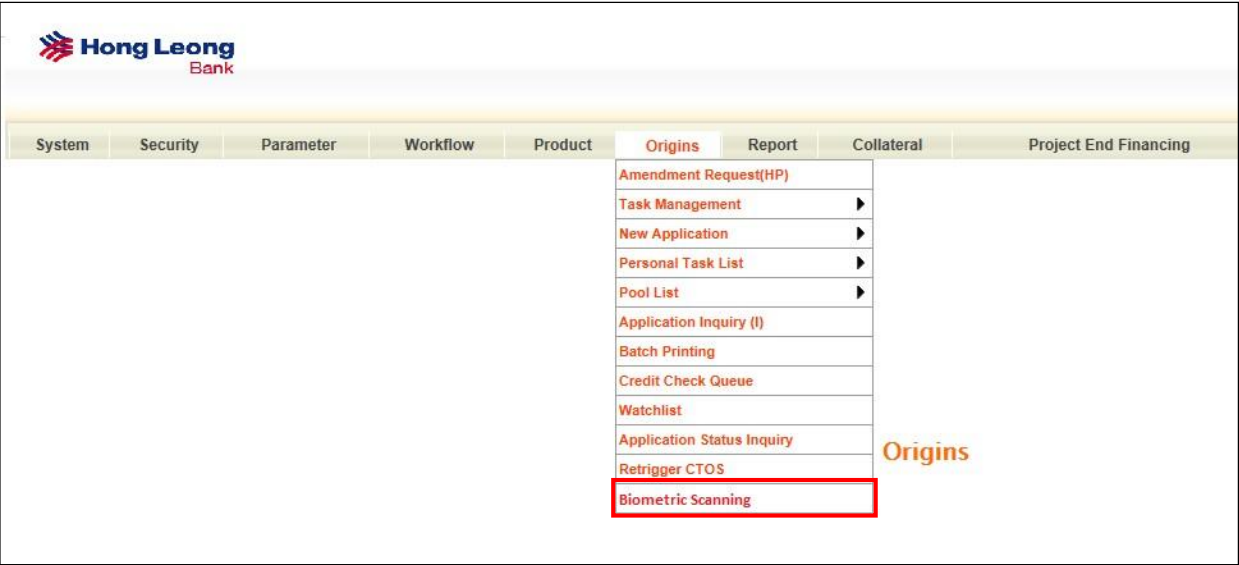
A simple template to display the required information as shown in Details Screen.

3. NEW SCREEN

3.1. MENU FOR BIOMETRIC SCANNING

The biometric screen will be made available under the following menu:

Note: User access matrix is applicable for this new menu.



3.2. BIOMETRIC SEARCH SCREEN

This screen will list down all Applicants and Sellers (Direct) which require to perform Biometric Scanning. The search criteria will be detailed out in the following sections:

Biometric Scanning
Origins version 6.2.15.0 build 1373 - Biometric Scanning
SITLOADSHPMY01

admin1 | 24:44 [Logout](#)
Menu [Go](#)

11/01/2016 05:13 PM, Mon

System Security Parameter Workflow Product Origins Report Collateral Project End Financing

Biometric Scanning Inquiry Search

App. Ref. No.
ID No

[Search](#) [Clear](#)

App. Ref. No.	ID No	Name	Relationship	Biometric Status	Triggered Counter	Action
SGT/HP/2015/L0000662	880101-00-0004	ZAINIZAM HARON	Primary Applicant	Error - Time Out	4	
SGT/HP/2015/L0000664	880101-00-0004	ZAINIZAM HARON	Primary Applicant	Not Done		

1

Displaying 1 - 2 of 2

3.2.1 Field Definition for Biometric Search Screen

No	Column Title	Sort? (Y/N)	Sort Order	Icon/Button	Remarks
1.	App. Ref. No.	Y – Manual	Ascending		
2.	ID No.	Y – Manual			Based on Applicant / Seller ID Note: User can inquiry by ID No regardless ID1 or ID2 but the result will always display the ID1.
3.	Customer Name	Y – Manual			
4.	Relationship	Y – Manual			Applicant Type (i.e Primary Applicant, Guarantor, Owner) -OR- Seller (for Direct Seller)
5.	Biometric Status	Y – Manual			Valid Value: 1. Matched 2. Unmatched 3. Error (with description) 4. Not Done
6.	Triggered Counter	Y – Manual			The number of Biometric Scanning attempt (including Error)
7.	Action	N			Click this icon to go Biometric Scanning Details Screen

3.2.2 Search Application or Applicant/Seller for Biometric Scanning

Function	Search Applicant / Seller for Biometric Scanning
Goal	To list all applicants / sellers (direct) which require to perform Biometric Scanning
Primary Actor/ Secondary Actor	All users
Trigger	1. By App. Ref. No. 2. By Customer ID No / Seller ID No Click 'Search' button. (Note: system not allows to Search All).

Main Course of Events	From the Biometric Search Screen when user clicks 'Search' button, system will check the following: 1. Table Matrix for the indicator to enable the Biometric Scanning (refer to item 5). If met the criteria, system will display the applicants. 2. General Constant for BIOMETRIC_CHECK_SELLER to identify if the Seller (Direct) is require for Biometric scanning. If the indicator is turn ON, system will display the seller. Else, display blank. The table listing will based on the draft above. Note: Biometric Search screen doesn't function like a normal Application Inquiry screen where the searching logic is based on applicant level. Hence, there is no handling to filter the application by Sales ID.
Alternative	-

3.3. BIOMETRIC SCANNING DETAILS SCREEN



Biometric Scanning Details

Origins version 6.2.15.0 build 1373 - Biometric Scanning Details
SITLOADSHPMY01

11/01/2016 05:05 PM, Mon

SystemSecurityParameterWorkflowProductOriginsReportCollateralProject End Financing

Biometric Scanning Details

Staff ID

03229

Date & Time Request

22/12/2015 05:23:39

Date & Time Response

22/12/2015 05:23:39

Biometric Status

Error - Time Out

Triggering Point

Host Error

Triggering Counter

1

Remarks

Biometric Scan

Report

Save

Back

3.3.1 Field Definition for Biometric Scanning Details Screen

No	Column Title	Sort? (Y/N)	Sort Order	Icon/Button	Remarks
1.	Staff ID	N			- Staff ID which is currently logged in to perform the biometric scan
2.	Date & Time Request	N			- Upon trigger for authentication - Refer to the latest record for multiple triggering
3.	Date & Time Response	N			- Upon received authentication result - Refer to the latest record for multiple triggering
4.	Biometric Status	N			Matched / Unmatched / Error
5.	Triggering Point	N			- Current workflow step when user trigger the biometric scanning - If the case has multiple current steps (ie. split workflow), display all steps. - Refer to the latest record for multiple triggering
6.	Triggering Counter	N			Number of Biometric Scanning attempt (including Error).

7.	Remarks	N			Free text format for User to input the remarks. There will be a validator to check this Remarks field (refer to item 6.2)
8.		N		Biometric Scan	When user click this button, the Biometric Verification Screen will be displayed (refer to item 4.5)
9.				Report	When user click this button, the report
No	Column Title	Sort? (Y/N)	Sort Order	Icon/Button	Remarks
					template will be downloaded (refer to item 8). The report will be generated per applicant.
10.				Back	When user click this button, it will go back to the Biometric Search Screen (refer to item 4.2)
11.				Save	To save the Remarks input by User

3.4. BIOMETRIC TAB (Within Application)

Application	Application Summary	Customer	Collateral	Facilities	Checklists	TV Check	Credit Check Result	Credit Review	Exposure Summary	FMU	Clause Generation	
Additional Credit Check	Biometric	Checker	CRA Defects	Maintenance	Disbursement	Other						

Select	Relationship	ID No	Name	Biometric Status
☒	Primary Applicant	661204-01-5035	Teh Boon Chun	Not Done
☐	Guarantor	670520-14-5100	Chin Chai Lim	Not Done

Displaying 1 - 2 of 2

Biometric Scanning Details

Staff ID
Date & Time Request
Date & Time Response
Biometric Status Not Done
Triggering Point
Triggering Counter
Remarks

Report Save

3.4.1 Field Definition for Biometric Tab

No	Column Title	Sort? (Y/N)	Sort Order	Icon/ Button	Remarks
Record Selection					
1.	Select	N		☒	Tick to view the details for this record Note: System will default to select the first record.
2.	Applicant Type	N			
3.	ID No.	N			
4.	Name	N			
Biometric Scanning Details					
5.	Staff ID	N			- Staff ID which has performed the Biometric

6.	Date & Time Request	N			- Date & Time Request of the Biometric Scan performed - Refer to the latest record for multiple triggering
7.	Date & Time Response	N			- Date & Time Response of the Biometric Scan performed - Refer to the latest record for multiple triggering
8.	Biometric Status	N			Matched / Unmatched / Error / Not Done
9.	Triggering Point	N			- Workflow Step which the Staff has performed the Biometric Scan - Refer to the latest record for multiple triggering
10.	Triggering Counter	N			Number of Biometric Scanning attempt (including Error)
No	Column Title	Sort? (Y/N)	Sort Order	Icon/ Button	Remarks
Record Selection					
11.	Remarks	N			Free text format for User to input the remarks. There will be a validator to check this Remarks field (refer to item 6.2)
12.				Report	When user click this button, the report template will be downloaded (refer to item 8). The report will be generated per applicant.
13.				Save	To save the Remarks input by User

Note: Save Button Access Control should be configured only for Sales at the Letter Printing Step so they may update the Remarks.

3.5. BIOMETRIC VERIFICATION SCREEN

This screen will be prompted upon User clicking the 'Biometric Scan' button (refer 4.3).

Hong Leong Bank

BIOMETRIC VERIFICATION
Origins version 5.2.15.0 build 1373 - BIOMETRIC VERIFICATION
ST/LOADSHIP/MS1

35592 | 38:38 Logout
Menu Go

11/01/2016 05:41 PM, Mon

System Security Workflow Origins Report Collateral Project End Financing

Biometric Device Selection:

☐ FID688 ☐ CSmarT680 ☒ SAGEM

Biometric Verification:

Name: Teh Boon Chun
ID: 661204015035
Status:

Verify

Back

Note: Details of Biometric Verification screen refer to **APPENDIX 1**

4. NEW MATRIX TABLES

4.1. MATRIX TABLE FOR BIOMETRIC SCANNING CHECKING CRITERIA

The matrix table below depicts the checking criteria to enable/disable the Biometric Scanning. Upon searching the application / applicants at Biometric Scanning Inquiry Search screen, system will check this matrix. If the criteria met and the indicator is 'YES', system will display the application / applicants in the result listing. Otherwise, system will not display which means the applicant is not require for Biometric Scanning.

Code: BIOMETRIC_CRITERIA_CHECK
Description: Criteria Checking to Enable / Disable Biometric Scanning for Applicant
Criteria: Customer Type, BNM Group Code, Relationship To Application, Id Type

		Primary Applicant		Guarantor		Owner	
		Identity Card - Blue	Identity Card - Red	Identity Card - Blue	Identity Card - Red	Identity Card - Blue	Identity Card - Red
Individual	Individual	YES	YES	YES	YES	YES	YES
	Commercial Bank	N/A	N/A	N/A	N/A	YES	YES
	Islamic Bank	N/A	N/A	N/A	N/A	YES	YES
	Investment bank	N/A	N/A	N/A	N/A	YES	YES
	International Islamic Bank	N/A	N/A	N/A	N/A	YES	YES
	Other Non - Bank FI	N/A	N/A	N/A	N/A	YES	YES
	Sole Proprietors	N/A	N/A	N/A	N/A	YES	YES
	Partnerships	N/A	N/A	N/A	N/A	YES	YES
	Professional Bodies	N/A	N/A	N/A	N/A	YES	YES
	Companies	N/A	N/A	N/A	N/A	YES	YES
	Federal/Central Government	N/A	N/A	N/A	N/A	YES	YES
	State Government	N/A	N/A	N/A	N/A	YES	YES

	Local Government	N/A	N/A	N/A	N/A	YES	YES
	Statutory Bodies	N/A	N/A	N/A	N/A	YES	YES
	Monetary Authority	N/A	N/A	N/A	N/A	YES	YES
	Trade Union	N/A	N/A	N/A	N/A	YES	YES

	Co-Operatives	N/A	N/A	N/A	N/A	YES	YES
	Societies/Associations	N/A	N/A	N/A	N/A	YES	YES
	International Organizations	N/A	N/A	N/A	N/A	YES	YES
	Stock Broking	N/A	N/A	N/A	N/A	YES	YES
	Others	N/A	N/A	N/A	N/A	YES	YES
Corporate	Individual	N/A	N/A	N/A	N/A	N/A	N/A
	Commercial Bank	N/A	N/A	N/A	N/A	N/A	N/A
	Islamic Bank	N/A	N/A	N/A	N/A	N/A	N/A
	Investment bank	N/A	N/A	N/A	N/A	N/A	N/A
	International Islamic Bank	N/A	N/A	N/A	N/A	N/A	N/A
	Other Non - Bank FI	N/A	N/A	N/A	N/A	N/A	N/A
	Sole Proprietors	N/A	N/A	N/A	N/A	N/A	N/A
	Partnerships	N/A	N/A	N/A	N/A	N/A	N/A
	Professional Bodies	N/A	N/A	N/A	N/A	N/A	N/A
	Companies	N/A	N/A	N/A	N/A	N/A	N/A
	Federal/Central Government	N/A	N/A	N/A	N/A	N/A	N/A
	State Government	N/A	N/A	N/A	N/A	N/A	N/A
	Local Government	N/A	N/A	N/A	N/A	N/A	N/A
	Statutory Bodies	N/A	N/A	N/A	N/A	N/A	N/A

	Monetary Authority	N/A	N/A	N/A	N/A	N/A	N/A
	Trade Union	N/A	N/A	N/A	N/A	N/A	N/A
	Co-Operatives	N/A	N/A	N/A	N/A	N/A	N/A
	Societies/Associations	N/A	N/A	N/A	N/A	N/A	N/A
	International Organizations	N/A	N/A	N/A	N/A	N/A	N/A
	Stock Broking	N/A	N/A	N/A	N/A	N/A	N/A
	Others	N/A	N/A	N/A	N/A	N/A	N/A

Notes:

1. A new common code will be created to define the ID Type for Biometric Scanning.
2. The matrix table will be made configurable.

4.2. MATRIX TABLE FOR MANDATORY CHECK

The matrix table below depicts the criteria for mandatory checking. There will be a workflow validator which refer to this matrix and if the criteria met and the indicator is “YES”, the Biometric Scanning is compulsory to be completed in order to pass the validation (refer to item 6.1). Otherwise, the Biometric Scanning is optional.

Code: BIOMETRIC_MANDATORY_CHECK
Description: Criteria check for Biometric Scanning validation
Criteria: BNM Group Code, Customer Type

	Individual			Corporate		
	Primary Applicant	Guarantor	Owner	Primary Applicant	Guarantor	Owner
Individual	YES	YES	YES	N/A	N/A	N/A
Commercial Bank	N/A	N/A	N/A	N/A	N/A	N/A
Islamic Bank	N/A	N/A	N/A	N/A	N/A	N/A
Investment bank	N/A	N/A	N/A	N/A	N/A	N/A
International Islamic Bank	N/A	N/A	N/A	N/A	N/A	N/A
Other Non - Bank FI	N/A	N/A	N/A	N/A	N/A	N/A
Sole Proprietors	N/A	N/A	N/A	N/A	N/A	N/A
Partnerships	N/A	N/A	N/A	N/A	N/A	N/A
Professional Bodies	N/A	N/A	N/A	N/A	N/A	N/A
Companies	N/A	N/A	N/A	N/A	N/A	N/A
Federal/Central Government	N/A	N/A	N/A	N/A	N/A	N/A
State Government	N/A	N/A	N/A	N/A	N/A	N/A
Local Government	N/A	N/A	N/A	N/A	N/A	N/A
Statutory Bodies	N/A	N/A	N/A	N/A	N/A	N/A
Monetary Authority	N/A	N/A	N/A	N/A	N/A	N/A
Trade Union	N/A	N/A	N/A	N/A	N/A	N/A
Co-Operatives	N/A	N/A	N/A	N/A	N/A	N/A
Societies/Associations	N/A	N/A	N/A	N/A	N/A	N/A
International Organizations	N/A	N/A	N/A	N/A	N/A	N/A
Stock Broking	N/A	N/A	N/A	N/A	N/A	N/A
Others	N/A	N/A	N/A	N/A	N/A	N/A

Notes:

1. The matrix table will be made configurable.
2. For application with multiple Owners, system require to check only ONE owner (this logic is not configurable).

3.

5. NEW WORKFLOW VALIDATOR & VALIDATION LOGIC

5.1. VALIDATOR FOR MANDATORY CHECK

A validator for mandatory check is required and will be configured in the workflow step after Approval Decision and before Host Interface. The checking logic for this validator as per follows in sequence:

1. System will check the mandatory setting in the Table Matrix (refer to item 5.2 above):
 - a. If the value is **“YES”**, system will continue check if the Biometric Scanning for that applicant (based on the criteria set in the matrix table) has been completed.
 - b. If not complete, system will prompt error message **“Please complete the Biometric scanning”** and the application will not able to proceed.
 - c. If complete, the application is able to proceed.
2. System will check the setup in general constant **BIOMETRIC_CHECK_SELLER**:
 - a. If the constant value = **I** (individual), system will check if the **Seller Type = Individual** and the Biometric Scanning for the **Seller (Direct)** has been completed.
 - b. If the constant value = **C** (corporate), system will check if the **Seller Type = Corporate** and the Biometric Scanning for the **Seller (Direct)** has been completed.
 - c. If not complete, system will prompt error message **“Please complete the Biometric scanning”** and the application will not able to proceed.
 - d. If complete, the application is able to proceed.

Details Seller Detail Market Value Checking

Seller Information

Seller Information

Seller Type ☒ Individual ☐ Corporate

Seller Name

ID Type / No. 1 Identity Card – Blue

ID Type / No. 2 --- select ---

Update Reset

Creation Date	Address Type	Address Line 1
No Record Found		

Add

5.2. VALIDATOR FOR UNSUCCESSFULL BIOMETRIC

A validator is required to check on the Remarks if Biometric Status = Unmatched or Error. This validator will be configured in the workflow at Letter Printing step. The checking logic for this validator as per follows in sequence:

1. System will check if the Biometric Status = Unmatched or Error (based on error code), then proceed to step 2. Else, validator check is pass.
2. System will check the Remarks field at Biometric Details Screen. If blank, system will stop the application from proceeding and prompt error message **“Biometric Status is Unmatched/Error without any remarks. Please input the remarks in the Biometric Details Screen.”**
3. Else, validator check is pass and application able to proceed.

5.3. VALIDATION FOR BIOMETRIC MAXIMUM RETRY

System requires to check the maximum retry for biometric attempt (including Error) upon triggering the Biometric as per follows:

1. Upon clicking the 'Biometric Scan' button, system will check the value set in general constant BIOMETRIC_ATTEMPT_COUNTER (refer to item 7.1) and if the number of attempt exceeded the value, system will prompt the error message **“Exceed maximum retry for biometric scanning”**.
2. Else, system will allow to proceed for biometric scanning.

5.4. VALIDATION FOR SUCCESSFULL BIOMETRIC

System requires to check the number of successful biometric attempt (excluding Error) upon triggering the Biometric as follows:

1. Upon clicking the 'Biometric Scan' button, system will check the Biometric log if any record with status = Matched or Unmatched.
2. System will compare the number of records against the value set in general constant BIOMETRIC_SUCCESSFUL_COUNTER (refer to item 7.2) and if the total records exceeded the value, system will prompt error message **“Exceed the number of successful attempt for biometric scanning”**.
3. Else, system will allow to proceed for biometric scanning.

4.

6. GENERAL CONSTANT

New general constant will be created to configure the following parameter:

1. Biometric Scanning Attempt Counter

To set the number of maximum attempt allowable for each customer regardless the biometric statuses (Matched / Unmatched / Error).

Business Type : Credit Evaluation Department
Constant Type : LOSA
Constant Code : BIOMETRIC_ATTEMPT_COUNTER
Code Description : Number of max retry for biometric scan **Constant Value** : 999

2. Biometric Scanning Successful Counter

To set the number of maximum retrigger allowable for each customer for successful biometric statuses only (Match / Unmatched).

Business Type : Credit Evaluation Department
Constant Type : LOSA
Constant Code : BIOMETRIC_SUCCESSFUL_COUNTER
Code Description : Number of max retrigger for successful attempt **Constant Value** : 999

3. Biometric Scanning for Seller (Direct)

The Biometric Scanning is applicable to the Seller (Direct) based on the applicant type set in this constant (i.e I, C) and the Biometric Scanning is mandatory to be completed.

Business Type : Credit Evaluation Department
Constant Type : LOSA
Constant Code : BIOMETRIC_CHECK_SELLER
Code Description : To enable / disable Biometric Scanning for Seller (Direct)
Constant Value : I (Individual), C (Corporate), OFF (disabled)

Notes:

1. If the Biometric Scanning is enable for the Seller (Direct), it is mandatory to be completed.
2. Seller Type = Corporate may not be applicable for biometric scanning.

3.

7. REPORT

A report template will be downloaded upon User clicking the “Report” button in the Biometric Scanning Details Screen (refer 4.3).

Sample report layout as per follows:

HONG LEONG BANK BERHAD / HONG LEONG ISLAMIC BANK BERHAD	
BIOMETRIC REPORT	
App. Ref. No.	INM/HP/2015/L0000506
Name	Test 5
ID No	920807-08-9982
Staff ID	03229
Date & Time Request	21/12/2015 02:42:36
Date & Time Response	21/12/2015 02:42:47
Biometric Status	Unmatched
Triggering Point	Full Data Entry and Dual Entry
Triggering Counter	1
Remarks	This remark is a free text. Thanks.

APPENDIX 1

Demo Screen Flow for LOAD\$ HP Biometric Verification

1. After click ‘Biometric Scan’ button, the following screen will be displayed:

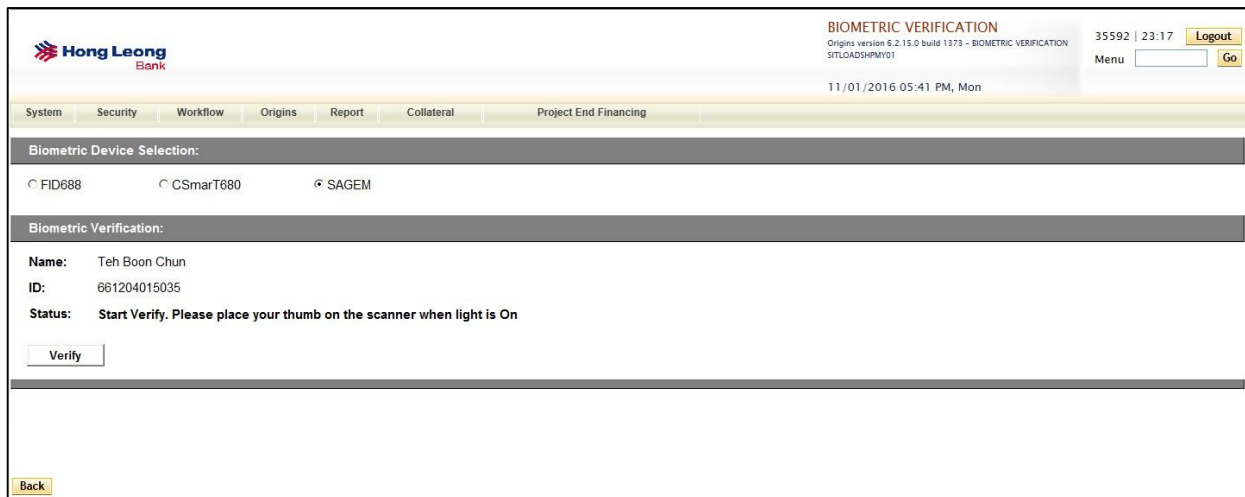
	BIOMETRIC VERIFICATION Origins version 5.2.15.0 build 1573 - BIOMETRIC VERIFICATION SIT/LOAD\$HP/15/01	35592 38:38 Logout
11/01/2016 05:41 PM, Mon		Menu Go
System Security Workflow Origins Report Collateral Project End Financing		
Biometric Device Selection:		
☐ FID688 ☐ CSmarT680 ☒ SAGEM		
Biometric Verification:		
Name: Teh Boon Chun		
ID: 661204015035		
Status:		
Verify		
Back		

2. Select the device (FID688 / CSmarT680 / SAGEM)

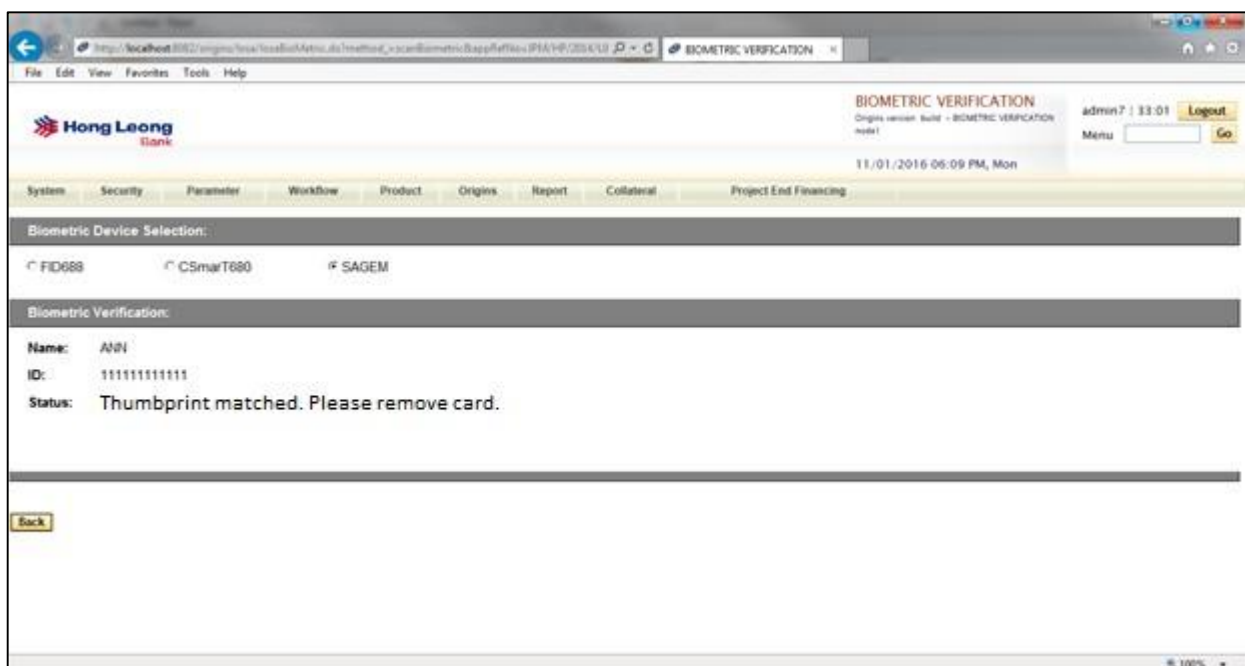
3. Click Verify button and an instruction message box will prompt out:



4. Insert MyKad into the Biometrics devices and click OK to continue the verifying process or Cancel to exit to the home screen.
5. Upon pressing OK to the message box, the following message will appear:



6. The device will start verifying the thumbprint with the information stored inside MyKad.
7. Upon successful verification the following message will appears:-



8. Error message will be prompted if the verification failed as per below sample:

Notes:

- (a) The 'Verify' button will be disappeared after result returned from device and User only have option to click Back button.
- (b) Back button will save the result in DB and return to the LOAD\$ screen.

Please refer to **APPENDIX 2** for the list of error message available.

APPENDIX 2

Biometrics Status / Error Code Listing

No	Model	Status Code (Device)	Description (Device)	Status Code (LOAD\$)	Description (LOAD\$)
1	SmarTec (Integrity)	0	Success	0	Matched
		-2	Read MyKad Failed	-1	Error – Failed to read MyKad
		-3	Fingerprint Verification Failed	-2	Error – Fingerprint verification failed
		-4	COM Port Error	-3	Error – Time Out
		-11	Timeout Error	-3	
		248	Card Not Inserted	-3	
		20001	Card Not Removed From Device	-3	
		20004	User Quit to Proceed	-3	
		29000	Create Object Error	-3	
		Others	Unknown Error	-3	
		20002	Error - IC Number Not Tally	-4	Unmatched

		20003	SmartTec Serial No Not Valid	-5	Error – Device Serial Number Not Valid
2	Sagem	0	Match Successful	0	Matched
		1	MyKad Not Found	-1	Error – Failed to read MyKad
		2	Software Dongle Not Found	-2	Error – Time Out
		3	Card Reader Not Found	-2	
		11	Matching Scanner Not Found	-2	
		12	Matching Software Dongle Not Found	-2	
		15	Match Timeout	-2	
		14	Match Invalid Template	-3	Error – Invalid template
		13	Error - Unmatched Thumbprint	-4	Unmatched
		20002	Error - IC Number Not Tally	-4	

- End of Document -